

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

Student Name

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file <FirstLast>_A07_GLMs.Rmd (replacing <FirstLast> with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
chem_physics <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"), stringsAsFactors = TRUE)
```

```
chem_physics$sampleddate <- as.Date(chem_physics$sampleddate, format = "%m/%d/%y")
```

```
str(chem_physics$sampleddate)
```

```
## Date[1:38614], format: "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" ...
```

```
head(chem_physics$sampleddate)
```

```
## [1] "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27"
```

```
## [6] "1984-05-27"
```

```
#2
```

```
my_theme <- theme_minimal() +
```

```
  theme(
```

```
    # text
```

```
    plot.title = element_text(face = "bold", hjust = 0.5, size = 14),
```

```
    axis.title.y = element_text(size = 12, margin = margin(r = 8)),
```

```
    axis.text = element_text(size = 10),
```

```
    # lines & grids
```

```
    axis.ticks = element_line(linewidth = 0.3),
```

```
    panel.grid.major = element_line(linewidth = 0.2, linetype = "dashed", color = "grey75"),
```

```
    panel.grid.minor = element_blank(),
```

```
    # backgrounds
```

```
    plot.background = element_rect(fill = "white", color = NA),
```

```
    panel.background = element_rect(fill = "white", color = "grey85"),
```

```
    legend.key = element_rect(fill = "white", color = NA),
```

```
  )
```

```
# Set as default theme
```

```
theme_set(my_theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

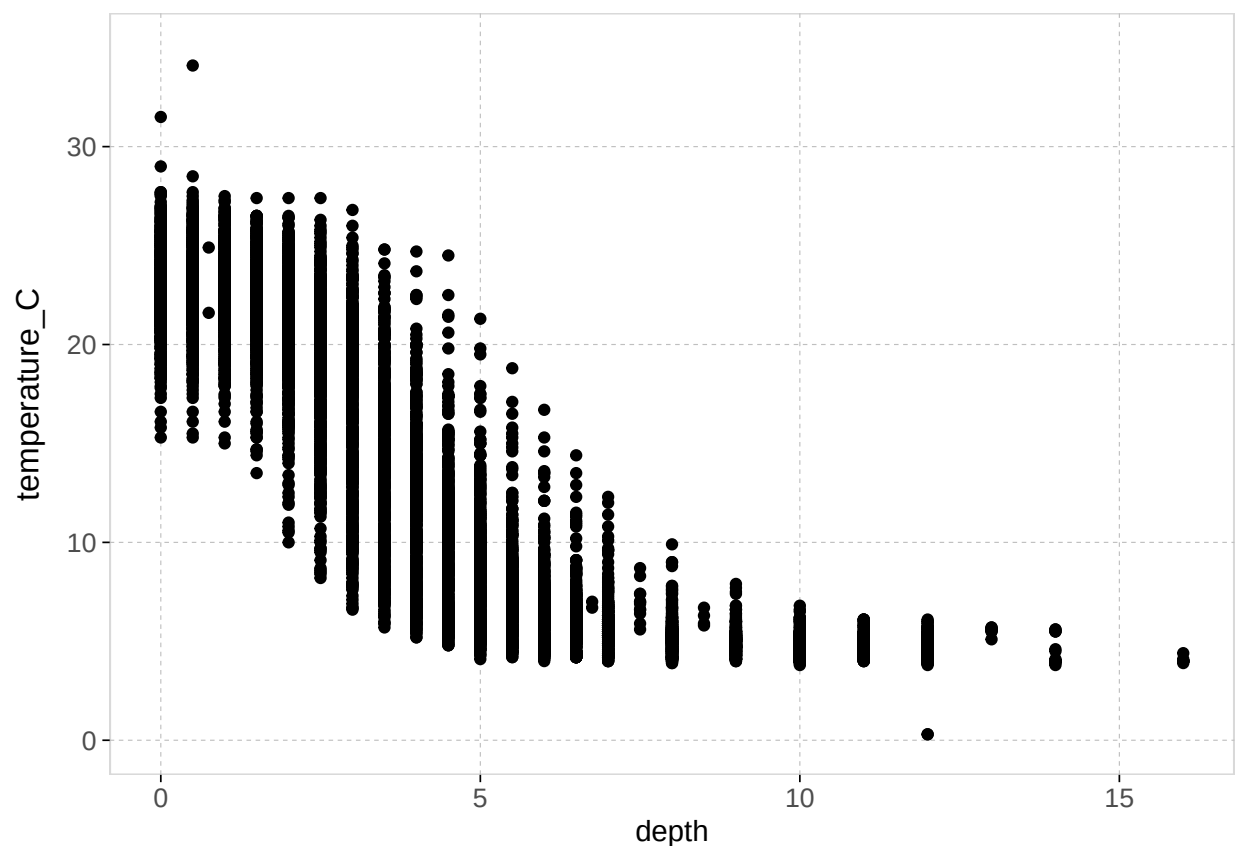
3. State the null and alternative hypotheses for this question: > Answer: H0: The mean lake temperature recorded during July did not change with depth across all lakes. Ha: The mean lake temperature recorded during July did change with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.

- Only the columns: lakename, year4, daynum, depth, temperature_C
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
chem_physics_july <- chem_physics %>%
  filter(format(sampledate, "%m") == "07") %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  drop_na()

#5
# Scatterplot
temptbydepth <-
  ggplot(chem_physics_july, aes(x = depth, y = temperature_C)) +
  ylim(0, 35) +
  geom_point() +
  my_theme
print(temptbydepth)
```

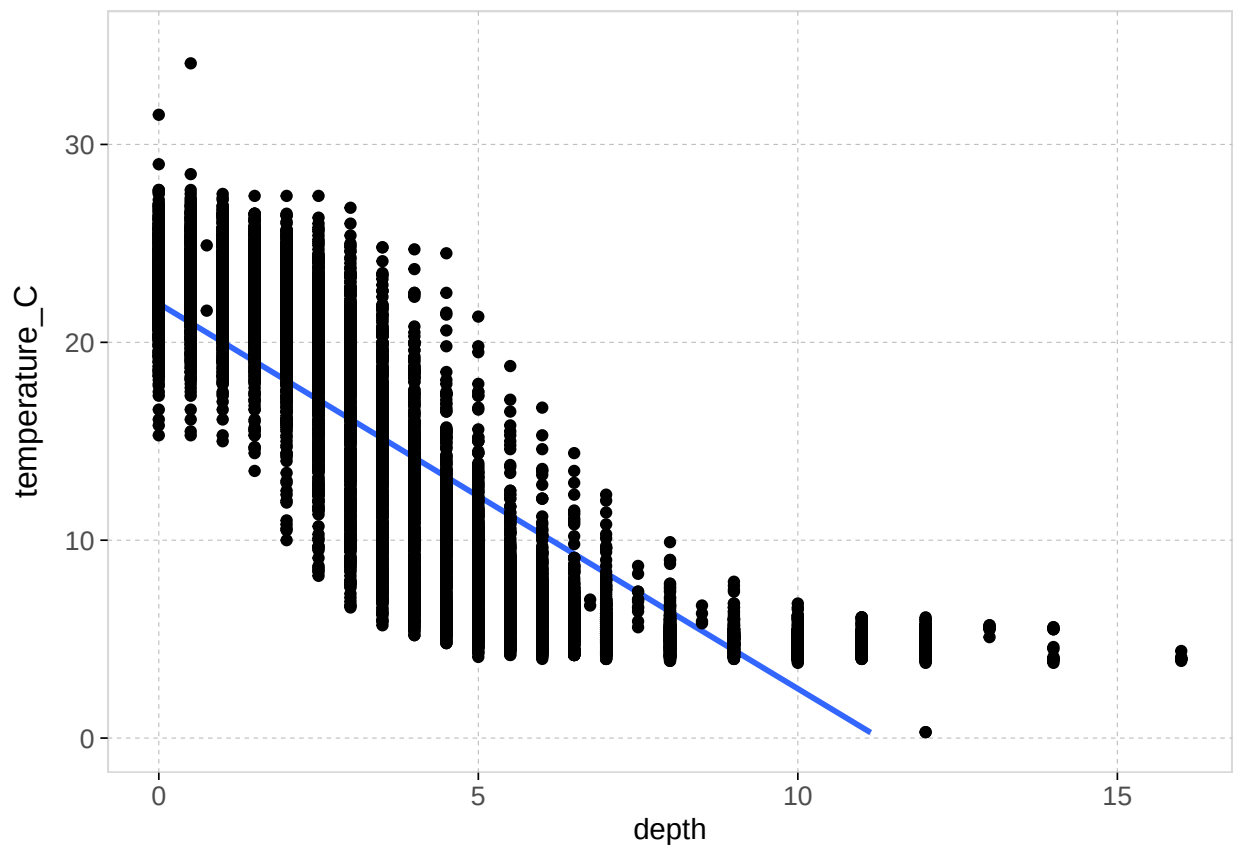


```
# Add a line and standard error for the linear regression
temptbydepth1 <-
  ggplot(chem_physics_july, aes(x = depth, y = temperature_C)) +
```

```
geom_smooth(method = "lm") +
ylim(0, 35) +
geom_point()
print(temptbydepth1)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

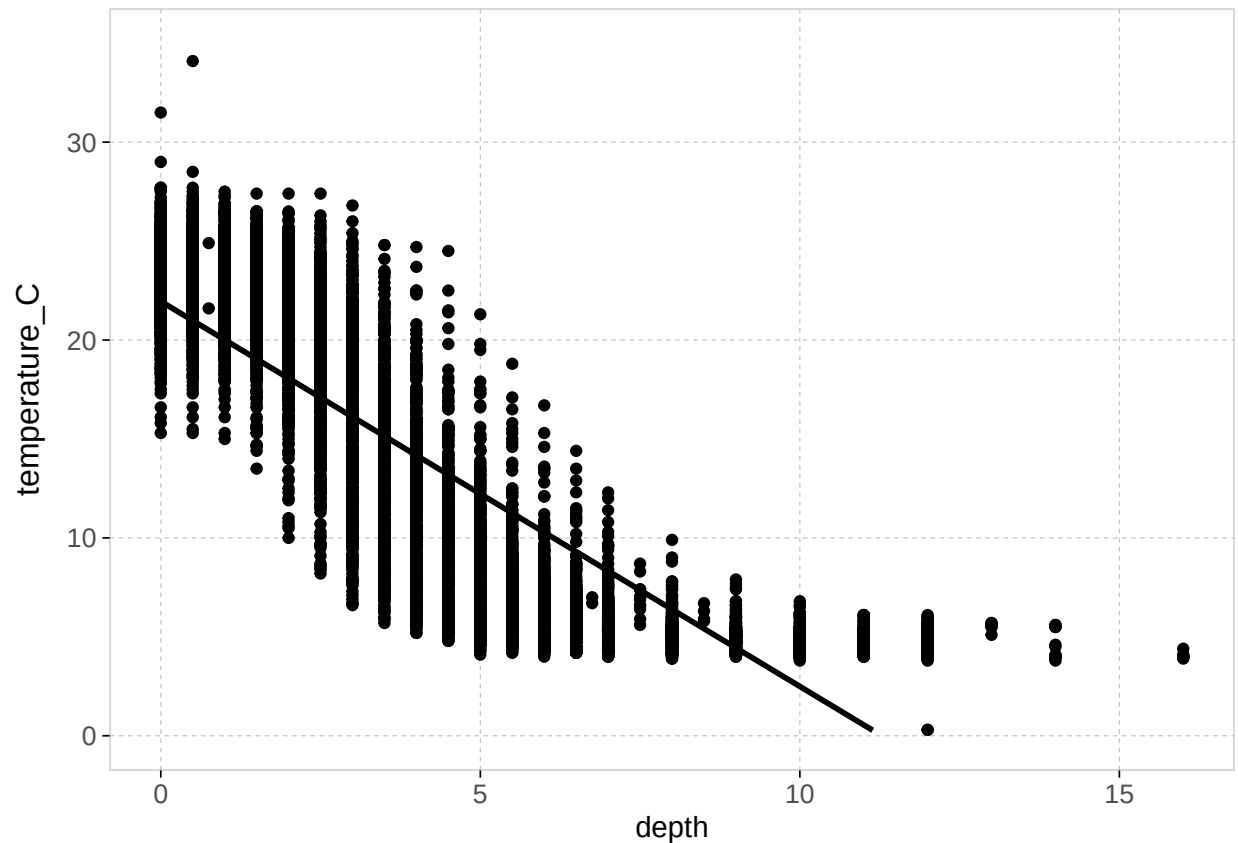
```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



```
# SE - confidence interval around smooth can also be removed
temptbydepth2 <-
  ggplot(chem_physics_july, aes(x = depth, y = temperature_C)) +
  ylim(0, 35) +
  geom_point() +
  geom_smooth(method = 'lm', se = FALSE, color = "black")
print(temptbydepth2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: Temperature decreases steadily with depth, showing a clear negative trend. The points mostly follow a straight line, though some curvature suggests natural layering in the lake.

7. Perform a linear regression to test the relationship and display the results.

```
#7
# Linear regression test
tempdepth_lm <- lm(data = chem_physics_july, temperature_C ~ depth)

# Display results
summary(tempdepth_lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = chem_physics_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173  -3.0192   0.0633   2.9365  13.5834
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth      -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The model shows a significant relationship between temperature and depth. About 73.87% of the variation in temperature is explained by changes in depth ($R^2 \approx 0.7387$). Based on 9,726 degrees of freedom, this relationship is highly significant ($p < 2.2e-16$). Temperature is predicted to drop by about 1.95 °C for every 1 m increase in depth.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
tempdepth_AIC <- lm(data = chem_physics_july, temperature_C ~ year4 + daynum + depth)

#Choose a model by AIC in a Stepwise Algorithm
step(tempdepth_AIC) # none = 26066: lowest AIC when dropping none
```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4      1         101 141788 26070
## - daynum     1         1237 142924 26148
## - depth      1       404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = chem_physics_july)
```

```
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##      -8.57556      0.01134      0.03978     -1.94644

#10
tempdepth_AIC <- lm(data = chem_physics_july, temperature_C ~ year4 + daynum + depth)
summary(tempdepth_AIC)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = chem_physics_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994   0.32044
## year4        0.011345   0.004299   2.639   0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final model includes year4, daynum, and depth as predictors of temperature. Together they explain about 74.11% of the variation in temperature, showing a strong fit ($p < 2.2e-16$). This is only a slight improvement over the model with depth alone, suggesting depth is the main factor, with minor effects from year and day.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
# Format ANOVA as aov
temp_anova <- aov(temperature_C ~ lakename, data = chem_physics_july)
summary(temp_anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals   9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Format ANOVA as lm
```

```
temp_lm <- lm(temperature_C ~ lakename, data = chem_physics_july)
anova(temp_lm)
```

```
## Analysis of Variance Table
##
## Response: temperature_C
##              Df Sum Sq Mean Sq F value    Pr(>F)
## lakename      8  21642   2705.2  50.003 < 2.2e-16 ***
## Residuals   9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(temp_lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = chem_physics_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769   -6.614   -2.679    7.684   23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501   27.174 < 2e-16 ***
## lakenameCrampton Lake      -2.3145     0.7699   -3.006 0.002653 **
## lakenameEast Long Lake     -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake   -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake         -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake        -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake      -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake    -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

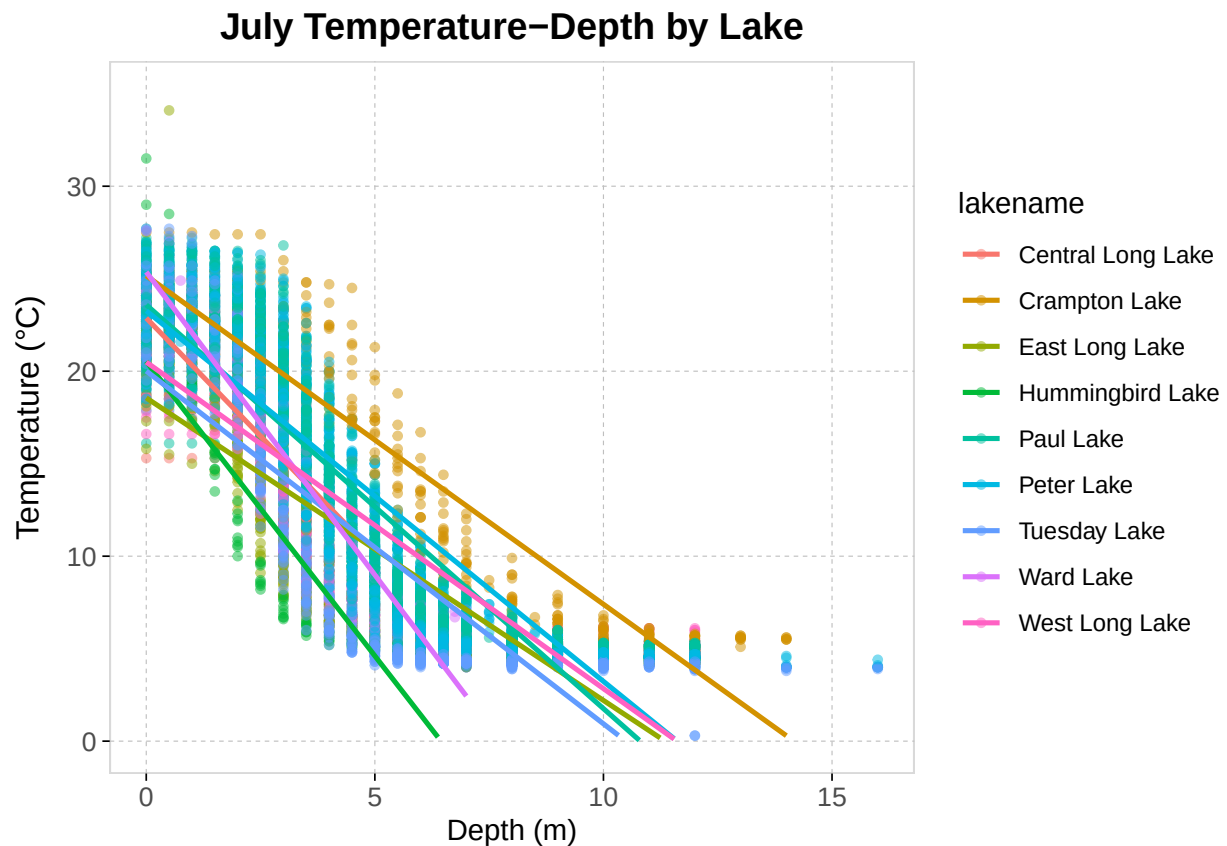
Answer: The ANOVA shows a significant difference in mean July temperature among lakes ($F(8, 9719) = 50.0, p < 2.2e-16$). The linear model indicates that several lakes, including East Long, Hummingbird, Tuesday, and West Long, are cooler than the reference lake, while others like Crampton and Paul are slightly cooler. Lake ID explains about 3.8% of the variation in temperature, which suggest that local lake characteristics influence water temperatures in July.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
lake_temp_plot <-
  ggplot(chem_physics_july, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5, size = 1.2) +
  geom_smooth(aes(group = lakename), method = "lm", se = FALSE, linewidth = 0.9) +
  ylim(0, 35) +
  labs(x = "Depth (m)", y = "Temperature (°C)",
       title = "July Temperature-Depth by Lake") +
  my_theme
lake_temp_plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```

#15
# Post-hoc test
# TukeyHSD() computes Tukey Honest Significant Differences
TukeyHSD(temp_anova)

## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = chem_physics_july)
##
## $lakename
##
## diff lwr upr p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913 0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake -0.8932661 -3.3684639 1.5819317 0.9714459
## West Long Lake-Crampton Lake -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake 0.5056106 -1.7364925 2.7477137 0.9988050
## Paul Lake-East Long Lake 3.5465903 2.6900206 4.4031601 0.0000000
## Peter Lake-East Long Lake 3.0485952 2.2005025 3.8966879 0.0000000
## Tuesday Lake-East Long Lake 0.8015604 -0.1363286 1.7394495 0.1657485
## Ward Lake-East Long Lake 4.1909554 1.9488523 6.4330585 0.0000002
## West Long Lake-East Long Lake 1.3109897 0.2885003 2.3334791 0.0022805
## Paul Lake-Hummingbird Lake 3.0409798 0.8765299 5.2054296 0.0004495
## Peter Lake-Hummingbird Lake 2.5429846 0.3818755 4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake 0.2959499 -1.9019508 2.4938505 0.9999752
## Ward Lake-Hummingbird Lake 3.6853448 0.6889874 6.6817022 0.0043297
## West Long Lake-Hummingbird Lake 0.8053791 -1.4299320 3.0406903 0.9717297
## Peter Lake-Paul Lake -0.4979952 -1.1120620 0.1160717 0.2241586
## Tuesday Lake-Paul Lake -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake 0.6443651 -1.5200848 2.8088149 0.9916978
## West Long Lake-Paul Lake -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake 1.1423602 -1.0187489 3.3034693 0.7827037
## West Long Lake-Peter Lake -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake 3.3893950 1.1914943 5.5872956 0.0000609
## West Long Lake-Tuesday Lake 0.5094292 -0.4121051 1.4309636 0.7374387
## West Long Lake-Ward Lake -2.8799657 -5.1152769 -0.6446546 0.0021080

# Extract groupings for pairwise relationships
temp_groups <- HSD.test(temp_anova, "lakename", group = TRUE)
temp_groups

```

```
## $statistics
##   MSerror   Df      Mean      CV
##   54.1016 9719 12.72087 57.82135
##
## $parameters
##   test   name.t ntr StudentizedRange alpha
##   Tukey lakename  9      4.387504  0.05
##
## $means
##               temperature_C      std      r      se Min  Max    Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake             13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake          11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake             14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake        11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##
##               Q75
## Central Long Lake 21.000
## Crampton Lake    22.300
## East Long Lake   15.925
## Hummingbird Lake 15.625
## Paul Lake        21.400
## Peter Lake        21.500
## Tuesday Lake     19.400
## Ward Lake        23.200
## West Long Lake   18.800
##
## $comparison
## NULL
##
## $groups
##               temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake          11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

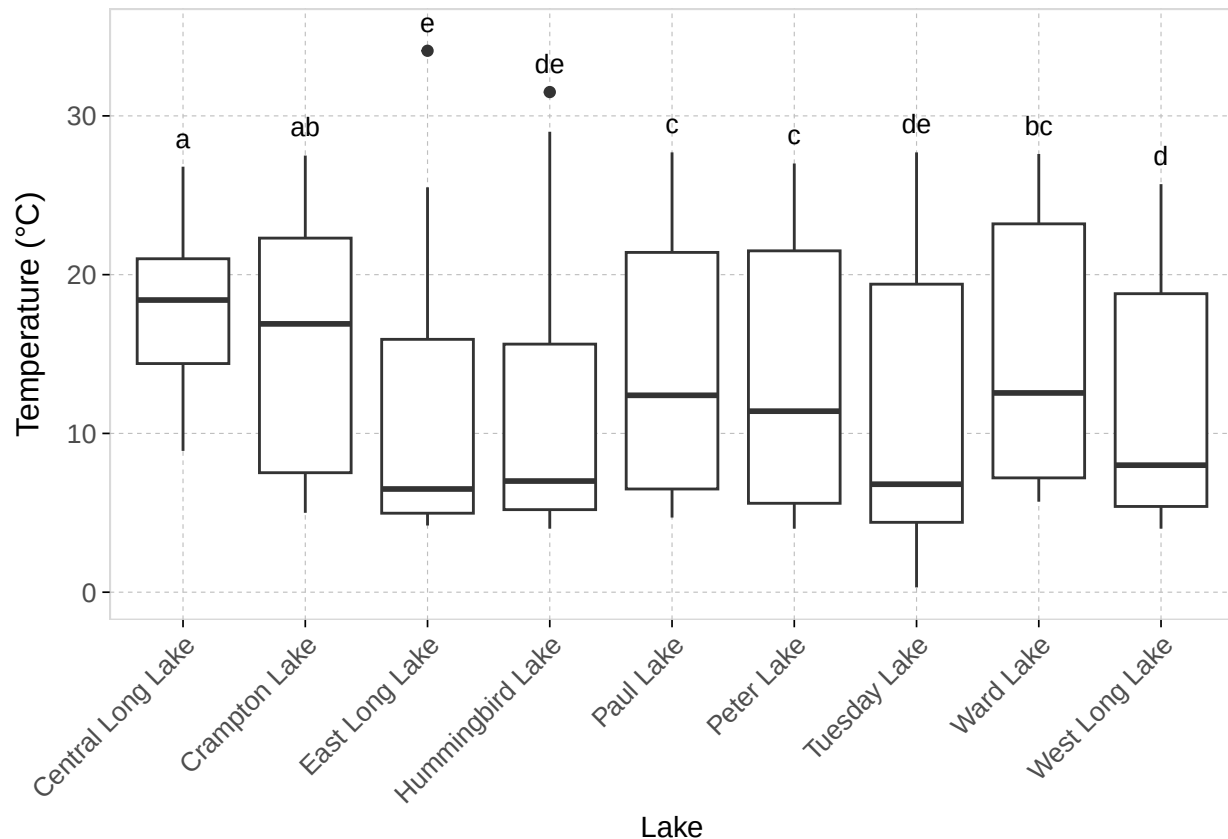
```
# Graph the results
```

```
tempbylake_boxplot <- ggplot(temp_anova, aes(x = lakename, y = temperature_C)) +
  geom_boxplot() +
  my_theme +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))+
```

```

stat_summary(geom = "text", fun = max, vjust = -1, size = 3.5,
             label = c("a", "ab", "e", "de", "c", "c",
                       "de", "bc", "d")) +
labs(x = "Lake", y = "Temperature (°C)") +
ylim(0, 35)
tempbylake_boxplot

```



16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Based on the Tukey HSD results, Peter Lake (group “c”) shares statistically similar mean temperatures with Paul Lake and Ward Lake, as they belong to overlapping groups that include the letter “c.” This means there is no significant difference in temperature among these three lakes. There is no lake that is statistically distinct from all the other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What’s another test we might explore to see whether they have distinct mean temperatures?

Answer: We can use two-sample t-test to directly test whether their mean temperatures differ significantly.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```

# Filter data for Crampton Lake and Ward Lake
cramptonvsward <- chem_physics_july %>%
  filter(lakename == "Crampton Lake"|lakename == "Ward Lake")

#Format as a t-test
cramp_ward_t.test <- t.test(temperature_C ~ lakename, data = cramptonvsward)
cramp_ward_t.test

##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862

```

Answer: The t-test ($p = 0.26$) shows no significant difference in July mean temperature between Crampton Lake (15.35°C) and Ward Lake (14.46°C). Their means are statistically similar, matching the earlier Tukey HSD result that grouped them together.